

Qatar-1b

R-2201

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121005 · created 2026-08-14T00:45:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456205.7028 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.019
Transit depth (Rp/Rs)^2	2.01 +/- 0.53 [%]
Orbital Inclination (inc)	88.88 +/- 1.34
Airmass coefficient 1 (a1)	0.965 +/- 0.017
Airmass coefficient 2 (a2)	0.027 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	#2 - [345, 296]
Optimal Aperture	2.73
Optimal Annulus	7.05
Transit Duration (day)	0.0818 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.019 vs 0.1463 (deviation 0.23 σ)
Duration fit vs published	0.0818 d vs 0.0692 d (deviation 6.02 σ)
Mid-time O-C	0.2 min
Depth SNR	7.5
Residual scatter	1.72 %

Light curve

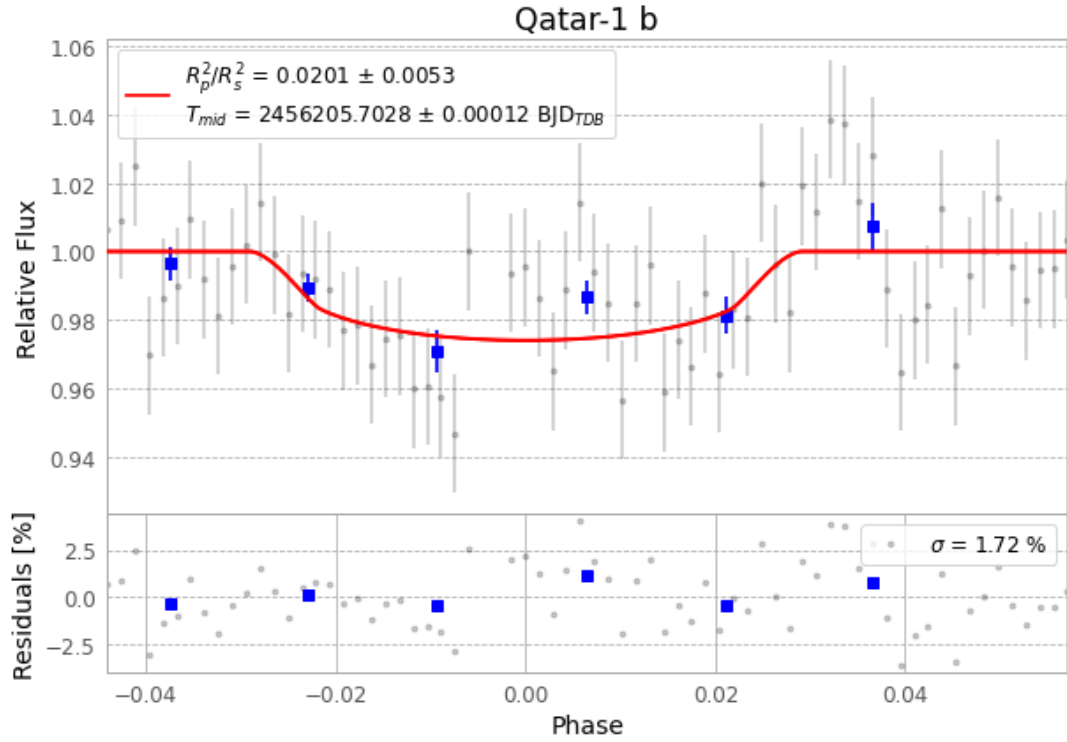


Figure 1 — FinalLightCurve_Qatar-1 b_2012-10-04.png (SHA-256 0b271e7f8d70082a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e41c449ce2fe0c74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), Qatar-1121005031812.FITS → Qatar-1121005064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121005.log (203d275a4844...), FinalLightCurve_Qatar-1 b_2012-10-04.png (0b271e7f8d70...), FinalLightCurve_Qatar-1 b_2012-10-04.csv (940a2cb08d88...)

Compute resources

Wall time 117.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:45Z.